The script analyzes a wild type and mutant strain of the p53 protein found in homo sapiens. This protein is attributed to being a transcription factor that serves as critical tumor suppressor that regulates cell division and combats cancer formation. The purpose of this analysis is to study how mutations in the p53 protein can lead to changes in the ability for the protein to bind to the DNA.

In this computational analysis, the wild type p53 being used has the PDB id of 3TS8 and the mutant p53 has the PDB id of 4MZR. The mutant p53 protein was engineered to have mutations on residues 121 and 122 changing the wild type serine and valine to phenylalanine and glycine respectively. To measure the effects of the mutations, the primary method was finding residues of both p53 proteins that were in contact with the DNA in their respective complex. This was accomplished by finding which residues were within 5 angstroms of the nearest DNA atom. This generated the same 14 residues on both the wild type and the mutant p53. However, only 13 residues had measurable changes that are listed below:

Residue	Wild type (Å)	Mutant (Å)	Change (Å)
239	3.92	4.24	+0.33
241	2.69	2.70	+0.02
248	3.69	3.37	-0.32
273	2.79	2.91	+0.12
274	4.48	4.74	+0.25
275	3.48	3.46	-0.02
276	3.00	2.78	-0.22
277	3.77	4.11	+0.34
280	3.25	2.83	-0.42
281	4.62	4.83	+0.21
283	3.37	2.63	-0.74
284	4.58	4.43	-0.16
288	4.97	4.54	-0.43

Improvements in contact are indicated by a negative change as this means that the distance between the residue and the DNA shortened (the greatest improvements indicated by green highlights). Inversely, positive changes indicate a disruption as the residue moves farther away from the DNA (greatest disruptions indicated by red highlights).

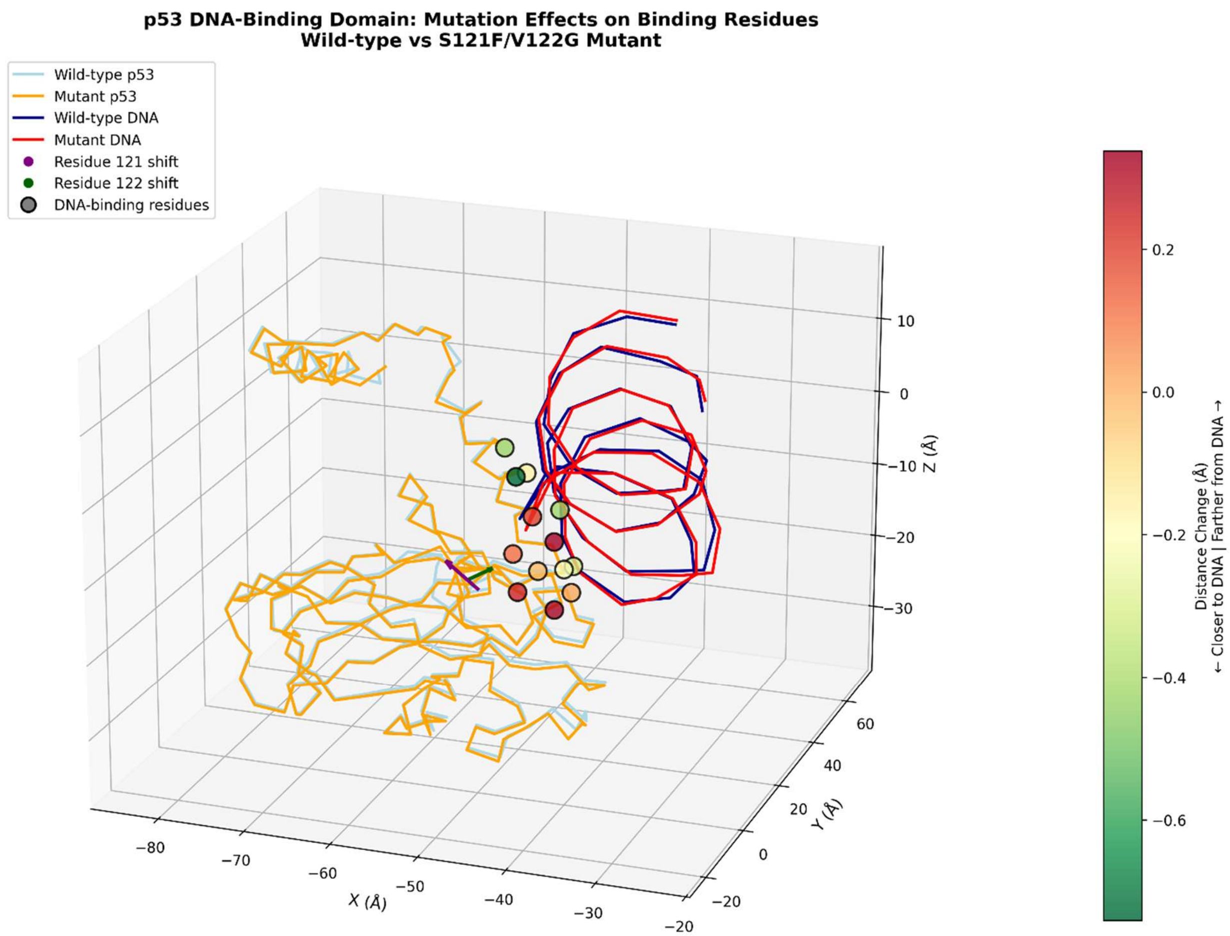

Figure 1- Model of both the wildtype and mutant DNA-p53 complexes

However, the importance of this analysis stems, not from the numbers, but how the changes in the protein-DNA complex affect the protein's ability to carry out biological function. 7 out of the 13 residues indicated above showed movement towards the DNA while the other 6 indicated disruptions in the binding residues. Despite, some of the residues showing "improvements" with regards to distance from then DNA, there is a high likelihood that these changes actually cause greater inability for the protein to appropriately perform its biological functions because the shifts across the protein create torsional strain stemming from the mutated residues that cascade along the entire structural body of the protein. This idea stems from the pattern of changes: as one moves downward along the protein (with respect to the z-axis), the residues begin shifting from being closer to the DNA to becoming farther from the DNA which is explained by the protein twisting because of the engineered mutations along the 121 and 122 residues. Since p53 is a transcription factor, its ability to bind precisely to specific sites on the DNA is critical for it initiate certain functions such as cell cycle arrest, DNA repair, or apoptosis. However, since the protein seems to be suffering torsional strain across the structural body, the active sites may not physically align their corresponding nucleotides causing reduced functionality.